Advancing Mental Health Problems with Machine Learning and Genetic Algorithms for Anxiety Classification in Bangladeshi University Students

Shahriar Siddique Ayon[1], Muhammad Ebrahim Hossain[1], Md Saef Ullah Miah[1](✉), M. Mostafizur Rahman[2], and Mufti Mahmud[3]

[1] Department of Computer Science, American International University-Bangladesh, Dhaka, 1229, Bangladesh {shahriarayon63, mdebrahim4200}@gmail.com, saef@aiub.edu

[2] Department of Mathematics, American International University-Bangladesh, Dhaka, 1229, Bangladesh mostafiz.math@aiub.edu

[3] Department of Computer Science, Nottingham Trent University, Nottingham NG1 4FQ, United Kingdom mufti.mahmud@ntu.ac.uk

Abstract. Mental health challenges, particularly anxiety, are a growing concern among university students in Bangladesh, impacting both their well-being and academic performance. This study aims to address this growing issue by developing a robust machine learning model tailored to classify anxiety levels among students. We employed eight well-known machine learning algorithms, with particular emphasis on a customized Genetic Algorithm (GA) optimized Logistic Regression (LR) model. The models were rigorously trained and evaluated using 5-fold cross-validation on a newly obtained mental health dataset that had never been investigated with such advanced approaches. This dataset contains information on 1,977 students from 15 of Bangladesh's leading universities and was thoroughly examined by five academics with 10 to 20 years of experience in academia and research. Our findings indicate that classic models, such as the Random Forest Classifier and Support Vector Classifier, attained accuracies of 92.68% and 96.72%, respectively. However, our proposed GA-based LR model succeeded them all, not only in accuracy but also in precision, recall, and F1-score, with an impressive 99.49% accuracy. The study also identified key anxiety-inducing factors, such as academic pressure and worry about academic affairs, providing valuable insights for targeted mental health interventions. These findings indicate the effectiveness of our customized GA-based ML model in enhancing mental health evaluations by identifying the fundamental causes of anxiety, providing vital insights to help Bangladeshi university students maintain their mental health.

Keywords: Mental Health Problems · Genetic Algorithm (GA) · Machine Learning (ML) · Bangladeshi University Students · Anxiety Classification.

1 Introduction

Mental health problems are a major global health concern, affecting millions of people of various ages and demographics. Among these issues, anxiety disorders are particularly prevalent, causing significant disruption in individuals' daily lives [21]. Anxiety is a common mental health issue characterized by excessive anxiety and fear that impairs daily functioning. It can take many forms, including generalized and social anxiety disorders, affecting both physical and psychological well-being [2, 13]. With the changing global scene, effective mental health treatments are urgently required, particularly for vulnerable groups. Among university students in Bangladesh, various mental disorders are emerging as significant public health concerns [15].

The rise in mental illness globally has been exacerbated by COVID-19, with factors such as gender, residence, relationship status, socioeconomic status, and loneliness significantly influencing mental health [10]. The pandemic has heightened anxiety and trauma, impacting mental health and increasing stress for students as universities shift to online learning [7]. These psychological effects have an immediate and long-term impact on pupils' energy levels, focus, mental ability, and academic success. University students in Bangladesh are particularly vulnerable to mental health issues such as anxiety as a result of academic stress and social expectations [11]. While prior research has identified these difficulties, more advanced methodologies are required to effectively classify and solve them.

While anxiety being a significant concern for university students in Bangladesh, there is a research gap in developing accurate, scalable methods for assessing anxiety levels. Traditional approaches often fail to detect subtle differences, leading to less effective interventions. This highlights the need for innovative strategies to improve classification accuracy and provide targeted support for students' mental health.

This research proposes a technique to improving mental health treatment by using machine learning and genetic algorithms to classify anxiety levels among Bangladeshi university students. Researchers have employed various ML algorithms, including random forests, support vector machines, and convolutional neural networks, to predict anxiety [1]. This study aims to compare the performance of eight popular ML algorithms and one customized genetic algorithm-based logistic regression model to predict anxiety among university students in Bangladesh. These advancements have the potential to significantly enhance support systems and serve as a model for similar research in other contexts.

The rest of this paper is organized as follows: Section 2 discusses relevant literature and previous studies. Section 3 discusses the dataset and experimental setup. Section 4 presents the results and their analysis. Section 5 explores the findings and their shared consequences. Finally, Section 6 summarizes the paper's findings and suggests areas for further investigation.

2 Literature review

WHO has recently designated mental health problems (MHPs) among university students as a global public health concern [19]. The intensity of depression, anxiety, and stress is evaluated using suggested cut-off scores [20]. The dataset uses standardized models to evaluate university students' mental health. Information gathered from 2028 students at 15 of Bangladesh's top-ranked universities, assessed using the GAD-7, PSS-10, and PHQ-9 models. These models may not fully capture all aspects of students' mental health issues. The study investigates the impact of sociodemographics on mental health and aims to provide data that clarifies the relationship between mental health and academic stress.

Post-secondary students have faced significant issues as a result of the COVID-19 pandemic, raising concerns regarding their psychological health examining the factors that influence the pandemic's prevalence of anxiety and sadness [22]. Meta-analyses using random effects revealed depression and anxiety rates of 30.6% and 28.2% among post-secondary students, respectively. Studies lacked diagnostic measures and couldn't explain heterogeneity factors. East Asian countries reported lower anxiety and depression rates.

Numerous frequent issues are shown by research on post-secondary students' mental health during COVID-19. Research indicates that students in higher-income nations frequently struggle with mental health disorders, unstable finances, physical health issues, food insecurity, social well-being, digital access, and housing issues [8]. One year after the epidemic, longitudinal surveys show that college students in China experienced an increase in suicidal thoughts and post-traumatic stress disorder (PTSD) but a decrease in anxiety and depression over time [6].

The COVID-19 pandemic has significantly impacted the mental health of Bangladeshi university students, leading to increased anxiety, depression, and overall distress. Machine learning and deep learning models, such as Siamese neural networks and Random Forest, have shown high accuracy in predicting mental health states. AI-based prediction tools can support early identification and intervention, aiding in the improvement of students' psychological well-being [3].

The Healthy Minds Study looked at the connection between college students' risk of suicide, mental health issues, and symptoms of despair and anxiety [5]. According to the study, if anxiety was also screened for, the percentage of suicidal thoughts detected by depression screening alone increased to 35% [17]. In addition, those who experienced anxiety but little sadness were more likely to attempt suicide; anxiety symptoms such as "feeling afraid something awful might happen" doubled the risk of having suicidal thoughts.

According to research, GAs can improve the accuracy of stress detection systems by selecting variables from physiological data, such as skin temperature and heart rate, more efficiently . When stress levels rise over predetermined thresholds, these algorithms enable real-time monitoring and alerting systems, enabling prompt treatments [12]. Furthermore, combining GAs with machine learning methods has demonstrated promise in enhancing stress detection predictive models, which raises the dependability of Internet of Things applications for mental health management [16]. There are still issues, though, such as the possibility of overfitting and the requirement for large datasets in order to train these algorithms efficiently [18].

3 Methodology

In our study, we followed a four-step approach: data collection and cleaning, feature engineering, selecting machine learning models, and optimizing performance using a genetic algorithm. Figure 1 concisely summarizes this process.

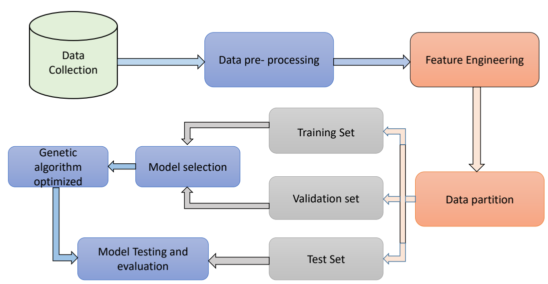


Fig. 1: Proposed methodology for multiclass anxiety classification.

3.1 Data Collection and Preprocessing

Our analysis uses a carefully produced dataset from the study "A Comprehensive Standardized Dataset on Mental Health Problems (MHPs) of University Students" [20]. This dataset contains information on 1977 students from Bangladesh's 15 top-ranked universities, which comprise 9 public and 6 private institutions. The data was thoroughly evaluated by five academics with 10 to 20 years of academic and research expertise. Three psychometric evaluation models were used systematically to assess anxiety, stress, and depression. For our investigation, we focused on anxiety in order to investigate the factors impacting students and find the major elements most closely connected with anxiety.

After collecting the data, we preprocessed the dataset by handling null values and encoding categorical variables through label and one-hot encoding. The original dataset had 1,977 data points and 16 columns, which expanded to 21 columns post-encoding. Key categorical variables included 'Age' (5 types), 'Gender' (3 types), 'University' (15 types), 'Department' (12 types), 'Academic Year' (5 types), and 'Current CGPA' (6 types). We also scaled the target variable, 'Anxiety Level', into four classes: 'Minimal Anxiety' (0), 'Mild Anxiety' (1), 'Moderate Anxiety' (2), and 'Severe Anxiety' (3). The data was then split into training (80%), testing (20%), and a validation subset from the training data (20%) to ensure balanced model preparation. The partitioned data for model training is shown in Table 1.

Table 1: Train, Test and Validation split of the employed dataset.

Target Colum	Number of Sample	Train	Test	Valid
Anxiety Level	1977	1265	395	317

3.2 Feature Engineering

After preprocessing, we moved to feature engineering. Feature engineering involves selecting the most relevant and impactful features from a dataset to improve model performance and clarity [4]. Figure 2 illustrates the feature importance for our dataset, derived using a linear regression model optimized with a genetic algorithm.

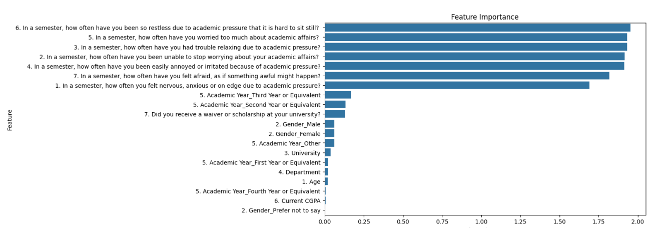


Fig. 2: Feature Importance Plot Highlighting the Impact of Features on Anxiety Levels.

There are a total of 20 features in the dataset, as shown by the feature importance in Figure 2. The first seven features have the greatest influence, followed by the next ten, which have a medium impact, and the final three, which have the least impact. The figure highlights that the top 3 features influencing anxiety levels are: '6. In a semester, how often have you been so restless due to academic pressure that it is hard to sit still?', '5. In a semester, how often have you worried too much about academic affairs?', and '3. In a semester, how often have you had trouble relaxing due to academic pressure?'. And the three least impactful features on anxiety levels were:'5. Academic Year_Fourth Year or Equivalent', '6. Current CGPA', '2. Gender_Prefer not to say'.

3.3 Machine Learning Model Selection

After preparing the data, we selected and implemented various machine learning models using Python libraries. We used a Decision Tree and a Random Forest classifier to optimize decision boundaries and improve accuracy. A Support Vector Classifier (SVC) found the best hyperplane for class separation, while a K-Nearest Neighbors (KNN) model predicted outcomes based on nearby instances. We employed Gaussian Naive Bayes (GaussianNB) for class probability estimation, LightGBM for efficient tree splitting, and a deep learning ANN to learn complex patterns using backpropagation and gradient descent. In our analysis, logistic regression (LR) was the best performer for multiclass anxiety classification. This multinomial LR extends binary logistic regression to handle multiple classes with the softmax function. This function calculates the probability of each class k as follows:

$$P(y = k \mid \mathbf{x}) = \frac{e^{\mathbf{w}_k^T \mathbf{x} + b_k}}{\sum_{j=1}^{K} e^{\mathbf{w}_j^T \mathbf{x} + b_j}} \tag{1}$$

Where, X represents the input feature vector, while W_k and b_k are the weight vector and bias for class K, respectively. The term $\mathbf{w}_k^T$ calculates the dot product of the weight vector and the input features. The denominator normalizes the probabilities across all K classes, enabling the model to predict the class with the highest probability, which is ideal for multi-class classification.

3.4 Optimizing ML Model with Genetic Algorithms

After assessing all machine learning models, logistic regression showed the best performance. To boost its accuracy, we applied a Genetic Algorithm (GA) to optimize the LR model. In this process, each individual in the population is evaluated using a fitness function that calculates the accuracy of the LR model with specific hyperparameters x.

$$Fitness(x) = Accuracy(LogisticRegression(x)) \tag{2}$$

Equation 2 evaluates each individual based on cross-validated accuracy. Next, tournament selection is used to identify the best individuals in the population based on their fitness. These selected individuals then undergo two-point crossover, where segments from two parent solutions are combined to create new offspring, expressed as follows:

$$Offspring_1, Offspring_2 = Crossover\left(Parent_1, Parent_2\right) \tag{3}$$

Equation 3 combines segments of parent solutions to create offspring. Mutation introduces random changes to the offspring to promote genetic diversity and explore new solutions, which may include randomly altering the hyperparameters. This process can be expressed as,

$$Mutant = Mutation\left(Offspring\right) \tag{4}$$

Equation 4 introduces random changes to maintain diversity. This process of selection, crossover, mutation, and evaluation is repeated over multiple generations, selecting individuals based on their fitness to converge on the optimal hyperparameters for improved model performance.

3.5 Hyper-parameter Tuning

After selecting the ML models, we fine-tuned the hyperparameters to improve performance. Table 2 shows the customized hyperparameter settings we used to maximize predictive efficiency across our models.

Table 2: Hyperparameter settings for different ML models utilized in this study.

Model	**Hyperparameter Settings**
Decision Tree Classifier	`random_state`: 42
Random Forest Classifier	`n_estimators`: 100, `max_depth`: 10, `random_state`: 42
Support Vector Classifier	`C`: 1.0, `kernel`: `rbf`, `gamma`: `scale`
K-Nearest Neighbors	`n_neighbors`: 5
Gaussian Naive Bayes	`var_smoothing`: 1e-9
LightGBM	`n_estimators`: 100, `learning_rate`: 0.1, `num_leaves`: 31
Artificial Neural Network	`neurons`: 128 (Layer 1), 64 (Layer 2), `Activation`: ReLU (Layer 1 & 2), Sigmoid (Output Layer), `Optimizer`: Adam, `Batch_Size`: 32, `Epochs`: 100
Logistic Regression	`C`: 0.1, `solver`: `lbfgs`, `max_iterations`: 1000
Proposed GA-Based Logistic Regression	`max_iterations`: 1000, `C`: Best value selected by GA, `solver`: Best value selected by GA (`liblinear`, `lbfgs`, `sag`, `saga`, `newton-cg`), `Population_size`: 50, `NGEN`: 10, `CXPB`: 0.5, `MUTPB`: 0.2

Table 2 summarizes the hyperparameter settings for each model to improve accuracy. The Decision Tree Classifier uses a 'random_state' of 42. The Random Forest Classifier has 100 estimators, a max depth of 10, and 'random_state' of 42. The Support Vector Classifier features 'C' set to 1.0, an 'rbf' kernel, and 'gamma' as 'scale'. KNN is set with 5 neighbors, and GaussianNB includes 'var_smoothing' of 1e-9. LightGBM uses 100 estimators, a learning rate of 0.1, and 31 leaves. The ANN consists of two layers with 128 and 64 neurons, ReLU and Sigmoid activations, Adam optimizer, batch size of 32, and is trained for 100 epochs. For LR, 'C' is set to 0.1, using the 'lbfgs' solver with a maximum of 1000 iterations for convergence. The Proposed GA-Based LR model is optimized using genetic algorithms with a maximum of 1000 iterations. The regularization parameter 'C' and the solver are selected based on the best values identified by the GA, with options including 'liblinear', 'lbfgs', 'sag', 'saga', and 'newton-cg'. For our model, a 'C' value of 0.3 and the 'newton-cg' solver were found to be the optimal choices for enhancing accuracy. The GA parameters are set with a population size of 50, 10 generations ('NGEN'), a crossover probability ('CXPB') of 0.5, and a mutation probability ('MUTPB') of 0.2 to refine the model's performance.

3.6 Evaluation Metrics

Once all settings were configured, the prediction model was assessed using key performance metrics for multi-class classification issues, such as accuracy, precision, recall, and F1-score. Each metric provides insights into the model's effectiveness, as represented by the following equations.

$$Accuracy = \frac{\sum_i TP_i}{\sum_i (TP_i + FP_i + FN_i)} \tag{5}$$

$$Precision_i = \frac{TP_i}{TP_i + FP_i} \tag{6}$$

$$Recall_i = \frac{TP_i}{TP_i + FN_i} \tag{7}$$

$$F1 - Score_i = 2 \times \frac{\text{Precision}_i \times \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \tag{8}$$

Where TP_i is the number of true positives for class i, FP_i is the number of false positives for class i, and FN_i is the number of false negatives for class i. And in multi-class classification, these metrics can be combined across classes using various averaging methods, including micro, macro, and weighted averages.

The Area Under the ROC Curve (AUC) is a key metric in ML, with values above 0.5 indicating good performance. In multi-class classification, performance is summarized using macro-average AUC, which treats all classes equally, and weighted-average AUC, which accounts for class imbalances.

4 Experimental Results and Analysis

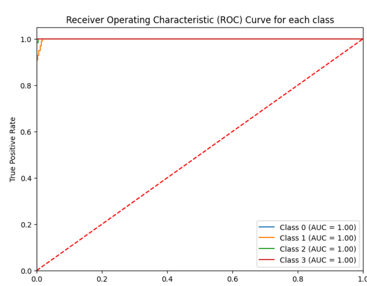
In this study, we developed ML models on Google Colab, benefiting from its efficient environment and collaboration features. All code was implemented in Python3 with Scikit-Learn, and we utilized the DEAP library for genetic algorithms, which offers a framework for individuals, populations, fitness functions, and genetic operators. This section compares prediction outcomes for several custom models. In this analysis, we examine the results of each model, with the final prediction outcomes on the testing dataset summarized in Table 3.

The performance comparison of various models reveals notable differences in precision, recall, F1-score, and accuracy, as shown in Table 3. The Random Forest model delivered strong results with 92.68% accuracy, while Decision Tree and KNN models showed moderate performance, with accuracies of 86.11% and

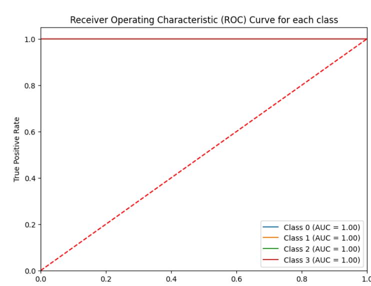
Table 3: Comparison of Model Performance Metrics.

Model	Precision (Weighted)	Recall (Weighted)	F1-Score (Weighted)	Accuracy (%)
Decision Tree	0.8632	0.8611	0.8607	86.11
Random Forest	0.9266	0.9268	0.9266	92.68
Support Vector Classifier	0.9683	0.9672	0.9672	96.72
K-Nearest Neighbors	0.8572	0.8485	0.8515	84.85
Gaussian Naive Bayes	0.9413	0.9394	0.9393	93.94
LightGBM	0.9519	0.9520	0.9520	95.20
Artificial Neural Network	0.5545	0.5957	0.5744	58.50
Logistic Regression	0.9852	0.9848	0.9848	98.48
Proposed GA-based Logistic Regression	**0.9950**	**0.9949**	**0.9949**	**99.49**

84.85%, respectively. The SVC and LightGBM models demonstrated high effectiveness, with weighted precision values of 0.9683 and 0.9519, and accuracy scores of 96.72% and 95.20%, respectively. Gaussian Naive Bayes also performed well, with 93.94% accuracy. In contrast, the ANN model under-performed, with significantly lower accuracy (58.50%) and other metrics. The Logistic Regression model showed strong results, with a weighted precision of 0.9852, recall of 0.9848, F1-score of 0.9848, and accuracy of 98.48%. The proposed GA-based Logistic Regression model outperformed all others, achieving outstanding performance metrics: a weighted precision of 0.9950, recall of 0.9949, F1-score of 0.9949, and accuracy of 99.49%.



(a) Logistic Regression AUC Curve.



(b) Proposed GA-based Logistic Regression AUC Curve.

Fig. 3: AUC Curves of the Top two Performing Models.

Figure 3 presents the AUC curves for the top 2 performing models in our multiclass classification with four classes. The AUC curves for our proposed GA-based logistic regression model 3b and logistic regression 3a are compared; both models achieve a perfect score of 1.0 for all classes. However, while the AUC curve in 3ainitially dips before reaching 1.0, the curve in 3b remains consistently at 1.0 throughout, indicating that our proposed model provides superior performance for this dataset.

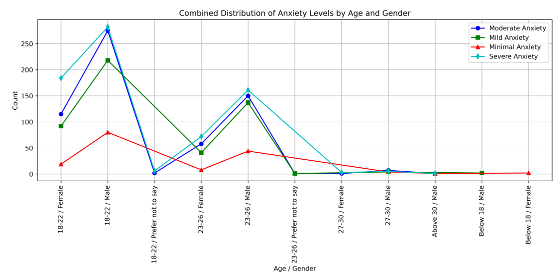


Fig. 4: Distribution of Anxiety Levels Across Four Classes by Age and Gender.

Figure 4 shows the distribution of anxiety levels across four classes by age and gender. The anxiety levels are as follows: Minimal Anxiety (7.99%), Mild Anxiety (25.04%), Moderate Anxiety (30.85%), and Severe Anxiety (36.14%). Age groups are: Below 18 (0.2%), 18-22 (64.39%), 23-26 (34.04%), 27-30 (1.21%), and Above 30 (0.15%). Gender distribution includes Male (69.39%), Female (30.10%), and Prefer Not to Say (0.51%). The graph highlights that Severe Anxiety is most common among 18-22 and 23-26 year-olds for both genders, while Moderate Anxiety is also prevalent in these age groups. Mild Anxiety is most frequent in 18-22 and 23-26 year-old males, with Minimal Anxiety being the least common across all demographics.

5 Discussion

Mental health issues, particularly anxiety, are a significant concern for university students in Bangladesh, impacting their well-being and academic performance [9, 14, 15]. This study aims to classify anxiety levels among these students to address this growing concern.

To achieve our study's objective, we used eight established ML algorithms alongside a proposed GA-based LR model. All models were trained with 5-fold cross-validation on a recent, unexplored dataset, with testing conducted on a separate test set. While previous studies identified factors influencing student anxiety, our analysis uncovered new contributing elements. The top three features impacting anxiety levels were: frequent restlessness due to academic pressure, excessive worry about academic affairs, and difficulty relaxing due to academic pressure. In our study, the majority of participants were Male (69.39%), followed by Female (30.10%) and a small percentage who Preferred Not to Say (0.51%), with ages ranging from under 18 to over 30. Severe Anxiety was most common among males aged 18-26 and females aged 18-22, while Moderate Anxiety primarily affected the same age groups. This suggests that the highest anxiety levels are found in males aged 18-26 and females aged 18-22.

Previous studies achieved strong results using RFC and SVC models. In our study, RFC and SVC models showed high accuracies of 92.68% and 96.72%, respectively, but our LR model surpassed them with 98.48% accuracy. By optimizing LR with a genetic algorithm, we achieved an outstanding accuracy of 99.49%, along with precision, recall, and F1-scores of 0.9950, 0.9949, and 0.9949, respectively, and a perfect AUC of 1.00. This GA-based LR model shows promise for predicting mental health issues, including anxiety, which is a major concern in Bangladesh.

6 Conclusion and Future Work

This study demonstrates that the GA-optimized LR model accurately classifies anxiety levels among Bangladeshi university students. The model surpasses standard methods in accuracy, precision, recall, and F1-scores due to the use of evolutionary algorithms for hyper-parameter optimization. The findings also identify critical elements that contribute to anxiety, providing useful insights for mental health providers and policymakers. Also this study enhances our understanding of anxiety predictors and lays a strong foundation for future mental health classification, underscoring the potential of machine learning to address critical public health challenges.

Future studies could expand the GA-based LR model to address other mental health issues beyond anxiety. Incorporating diverse datasets and exploring alternative evolutionary techniques may enhance the model's generalizability and predictive ability. Additionally, integrating Explainable AI (XAI) could clarify the model's decision-making, helping mental health professionals understand anxiety predictors. Collaborating with mental health specialists to implement these models in real-world settings will be crucial for improving mental health outcomes.

Data and Code Availability

The codes and data employed in this study can be found in the following GitHub repository. https://github.com/ShahriarAyon63/Anxiety_genetic_algorithm

References

1. Abd Rahman, R., Omar, K., Noah, S.A.M., Danuri, M.S.N.M., Al-Garadi, M.A.: Application of machine learning methods in mental health detection: a systematic review. Ieee Access **8**, 183952–183964 (2020)

2. Ahmed, R., Fahad, N., Miah, M.S.U., Hossen, M.J., Morol, M.K., Mahmud, M., Rahman, M.M.: A novel integrated logistic regression model enhanced with recursive feature elimination and explainable artificial intelligence for dementia prediction. Healthcare Analytics **6**, 100362 (2024)
3. Atique, M.M.A.B., Bappi, M.I., Kim, K., Choi, K., Ahamad, M.M., Reza, K.M.: Impact of covid-19 on bangladeshi university students mental health: Ml and dl analysis. medRxiv pp. 2024–05 (2024)
4. Ayon, S.S., Hossain, M.E., Rabbi, A., Saef Ullah Miah, M., Rahman, M.M.: Predictive modeling and early detection of white spot disease in shrimp farming using machine learning: A case study in bangladesh. In: International Conference on Trends in Electronics and Health Informatics. pp. 263–273. Springer (2023)
5. Casey, S.M., Varela, A., Marriott, J.P., Coleman, C.M., Harlow, B.L.: The influence of diagnosed mental health conditions and symptoms of depression and/or anxiety on suicide ideation, plan, and attempt among college students: Findings from the healthy minds study, 2018–2019. Journal of affective disorders **298**, 464–471 (2022)
6. Chen, Y., Ke, X., Liu, J., Du, J., Zhang, J., Jiang, X., Zhou, T., Xiao, X.: Trends and factors influencing the mental health of college students in the post-pandemic: four consecutive cross-sectional surveys. Frontiers in psychology **15**, 1387983 (2024)
7. Copeland, W.E., McGinnis, E., Bai, Y., Adams, Z., Nardone, H., Devadanam, V., Rettew, J., Hudziak, J.J.: Impact of covid-19 pandemic on college student mental health and wellness. Journal of the American Academy of Child & Adolescent Psychiatry **60**(1), 134–141 (2021)
8. Dey, P., De Souza, L.R.: Public health challenges for post-secondary students during covid-19: A scoping review. Community Health Equity Research & Policy p. 2752535X241257561 (2024)
9. Hossain, M.J., Ahmmed, F., Khandokar, L., Rahman, S.A., Hridoy, A., Ripa, F.A., Emran, T.B., Islam, M.R., Mitra, S., Alam, M.: Status of psychological health of students following the extended university closure in bangladesh: results from a web-based cross-sectional study. PLOS Global Public Health **2**(3), e0000315 (2022)
10. Hossain, M.M., Tasnim, S., Sultana, A., Faizah, F., Mazumder, H., Zou, L., McKyer, E.L.J., Ahmed, H.U., Ma, P.: Epidemiology of mental health problems in covid-19: a review. F1000Research **9** (2020)
11. Islam, M.A., Barna, S.D., Raihan, H., Khan, M.N.A., Hossain, M.T.: Depression and anxiety among university students during the covid-19 pandemic in bangladesh: A web-based cross-sectional survey. PloS one **15**(8), e0238162 (2020)
12. Jain, R.K., Jain, S.K., Vasal, S.: Using iot optimize different psychological problems in human brain with the help of genetic algorithm. In: 2024 IEEE 13th International Conference on Communication Systems and Network Technologies (CSNT). pp. 574–578. IEEE (2024)
13. Kenwood, M.M., Kalin, N.H., Barbas, H.: The prefrontal cortex, pathological anxiety, and anxiety disorders. Neuropsychopharmacology **47**(1), 260–275 (2022)
14. Khan, M.N.A., Miah, M.S.U., Shahjalal, M., Sarwar, T.B., Rokon, M.S.: Predicting young imposter syndrome using ensemble learning. Complexity **2022**(1), 8306473 (2022)
15. Mamun, M.A., Hossain, M.S., Griffiths, M.D.: Mental health problems and associated predictors among bangladeshi students. International Journal of Mental Health and Addiction **20**(2), 657–671 (2022)
16. Mokhayeri, F., Akbarzadeh-T, M., Toosizadeh, S.: Mental stress detection using physiological signals based on soft computing techniques. In: 2011 18th Iranian Conference of Biomedical Engineering (ICBME). pp. 232–237. IEEE (2011)

17. Moskow, D.M., Lipson, S.K., Tompson, M.C.: Anxiety and suicidality in the college student population. Journal of American College Health (2022). https://doi.org/10.1080/07448481.2022.2060042
18. Sandeep, N.L.: Human stress detection based on sleeping habits through machine learning algorithms. Indian Scientific Journal Of Research In Engineering And Management (2024). https://doi.org/10.55041/ijsrem31628
19. Syeed, M., Hossain, M.S., Rahman, A., Fatema, K., Khan, R.H., Uddin, M.F.: Mental health assessment of university students during and after the covid pandemic: A comparative study. Mental Health Assessment of University Students during and after the COVID Pandemic: A Comparative Study (January 17, 2024) (2024)
20. Syeeda, M.M., Rahmana, A., Akterc, L., Fatemaa, K., Khana, R.H., Karima, M.R., Hossaina, M.S., Uddina, M.F.: A comprehensive standardized dataset on mental health problems (mhps) of university students. measurement **8**, 9 (2024)
21. Szuhany, K.L., Simon, N.M.: Anxiety disorders: a review. Jama **328**(24), 2431–2445 (2022)
22. Zhu, J., Racine, N., Xie, E.B., Park, J., Watt, J., Eirich, R., Dobson, K., Madigan, S.: Post-secondary student mental health during covid-19: a meta-analysis. Frontiers in Psychiatry **12**, 777251 (2021)